🚀 Olist Capstone Project: Final Report & Recommendations

Executive Summary

This analysis was commissioned to investigate the root cause of a high volume of 1-and-2-star customer reviews on the Olist platform. The primary business hypothesis was that poor logistics performance—specifically, long delivery times—was the main driver of customer dissatisfaction.

This analysis confirms the hypothesis is correct.

After processing and analyzing 9 datasets, the findings show a direct, statistically significant, and negative correlation between delivery time and review score. The problem is concentrated in specific geographic regions and is driven by a distinct group of underperforming sellers. This report provides actionable recommendations to address these issues.

Key Findings (The "Why")

Our analysis of the data, visualized in the [Tableau Dashboard], revealed four key findings:

1. **The Link is Statistically Proven:** The visual correlation is not a coincidence. A Pearson correlation test performed in R confirmed a **statistically significant, strong negative correlation ($r = -0.976$, $p < 0.01$)** between average delivery time and customer review score.
2. **The Impact is Quantifiable:** The Delivery Time vs. Review Score analysis shows that 1-star reviews are associated with an average delivery time of **21.2 days**, whereas 5-star reviews are associated with an average of **10.6 days**. This proves that every day of delay actively decreases customer satisfaction.
3. **Performance is Not Uniform by Region:** The geographic heatmaps (Seller Score by State and Delivery Days by State) clearly show that delivery performance and customer satisfaction are not uniform across Brazil. Specific states, particularly in the North and Northeast, show significantly longer delivery times and, as a direct result, lower average review scores.
4. **A Specific Seller Group is Responsible:** The Seller Performance Analysis scatter plot identifies a clear "problem quadrant." These are sellers who are both **slower than the platform average (14.9 days)** and have **review scores lower than the platform average (3.99 stars)**. This group is having a disproportionately negative impact on the platform's reputation.

Recommendations (The "Act")

Based on these data-driven findings, we recommend the following three-part action plan:

1. (Action: Seller Management) Launch a Seller QA & Performance Program

- **Recommendation:** Immediately engage the sellers identified in the "bottom-right" quadrant of the Seller Performance Analysis. These sellers require a formal process audit and retraining.
- **Justification:** This small group of sellers is your biggest liability. Focusing resources on them will yield the fastest improvement in platform-wide review scores.

2. (Action: Logistics) Audit Regional Logistics Partners

- **Recommendation:** Prioritize a performance review of the logistics carriers in the states identified by the maps as having the longest delivery times (e.g., the red-colored states).
- **Justification:** The data shows that your partners in these regions are failing. You must either renegotiate your service-level agreements (SLAs) or find new partners to service these areas effectively.

3. (Action: Customer Experience) Recalibrate Estimated Delivery Dates

- **Recommendation:** Use the state-by-state delivery averages we found in this analysis to create a more realistic and dynamic delivery estimation algorithm.
- **Justification:** The analysis proves customers are more frustrated by broken promises than by a long wait. It is better to promise delivery in 15 days and arrive in 14, than to promise 10 days and arrive in 12. This simple change will better manage customer expectations and immediately improve satisfaction.

Further Analysis: As a next step, I recommend performing text analysis (NLP) on the review_comment_message field to discover *why* customers are leaving bad reviews *besides* logistics (e.g., "damaged product," "wrong item").